Susie Shin

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U.S. Citizen

RESEARCH INTERESTS

- RF Engineering
- Integrated Circuit & PCB Design
- Aerospace Application

EDUCATION

Yonsei University — Seoul, Korea

- B.S. in Electrical and Electronic Engineering — Feb. 2027 (Expected)
  - Total GPA of 3.92/4.0, Major GPA of 3.95/4.0

University of California, San Diego — CA, USA

- Education Abroad Program (UCEAP) — Sep. 2025 – Jun. 2026 (Expected)

RESEARCH EXPERIENCE

- **Students for the Exploration and Development of Space (SEDS)** Lander Avionics Hardware — CA, USA
  [University of California, San Diego] — Oct. 2025 - present
  - **Ground Station / Communications**
    - Researched and developed a ground station communication system for remote monitoring during tethered lander tests
    - Worked on telemetry and emergency-stop communication links using 915 MHz and 2.4 GHz wireless systems
  - **Avionics / PCB Design**
    - Designed a PCB to interface the flight computer with GPS, IMU, and LiDAR modules using Altium Designer
    - Debugged and revised an existing schematic to resolve hardware integration issues
    - Designed a DC to DC converter for onboard power regulation
- **Microwave Integrated Circuit & System Lab (MICS)** [Yonsei University] — Seoul, Korea
  Advisor: Professor Byung-Wook Min — Jul. 2025 – Sep. 2025
  - **920-MHz RFID Helical Antenna**
    - Designed and fabricated a microstrip line for a 180-degree balun; optimized impedance matching and validated S-parameters using VNA
    - Utilized HFSS for circuit and EM simulation
  - **13-GHz In-band Full-duplex Radio**
    - Designed Cascaded Stages of Wilkinson Power Divider with Grounded Coplanar Waveguide using HFSS; analyzed insertion loss and isolation performance
    - Designed Layer Transition Line and Connector PCB Pattern using HFSS

PROJECTS

- **Ideal/Practical Operational Amplifier Design** [Yonsei University – Electronic Circuits 2] — Apr. / Jun. 2025
  - Differential amplifier, Current mirror, Negative feedback
- **Self-Driving Car Development** [Korea University Innovation Center for Engineering Education] — Feb. 2025
- **Full Wave Rectifier / Two-stage Amplifier Design** [Yonsei University – Electronic Circuits 1] — Dec. 2025
  - Diode connected transistor / Common emitter amplifier, Common drain amplifier
- **IR sensor PCB Design / Audio Equalizer Design** [Yonsei University – Analog Electronics Lab] — Nov. / Dec. 2025
  - Designed and fabricated an IR sensor PCB in Eagle CAD; validated proximity detection through circuit testing

- ◦ Built and tested a 3-band audio equalizer circuit incorporating active filters, audio mixer, and push-pull amplifier; analyzed frequency response

- **Integrated Control System for Drone Collision Prevention** (Selected Idea) — Jul. - Sep. 2024
  [NIA (National Information Society Agency), KOREN Research Cooperation Forum]
  - ◦ Designed and simulated an integrated control system to manage multiple drones through a centralized server for future drone industry applications
- **'Ear Care': Humidity-Based Sensor Design to Prevent Ear Infections from Wireless Earphones** — Aug. 2024
  [Samsung Electronics DS Division]
  - ◦ Multi-threshold humidity sensing system by connecting MOSFETs with different threshold voltages, exploiting changes in dielectric resistance due to moisture. Expected miniaturization and low-power design of the sensor.
- **'Aqua-Rescue': Vehicle Flood Detection and Response Model** (Excellence Award) — May. - Nov. 2023
  [Yonsei University, Nexon Korea]
  - ◦ Flood detection model for vehicles that senses external water levels and triggers a window-breaking device
  - ◦ Analog Circuit Design, Hardware Implementation, Mobile Application Development

EXTRACURRICULAR ACTIVITY

- **Students for the Exploration and Development of Space (SEDS)** [University of California, San Diego] — Oct. 2025
  - ◦ Lander Riptide, Avionics Hardware
- **Member, NAEK Young Engineers Honor Society (YEHS)** — Jul. 2025
  - ◦ Nationwide association of engineering students in Korea, operating under the National Academy of Engineering of Korea (NAEK)
- **Winter Convergence Program,** AI Object Recognition and Solar Tracking Robot Challenge — Jan. 2025
  [Korea University Innovation Center for Engineering Education]
- **Samsung Shining Star Program 5th** [Samsung Electronics DS Division] — Aug. 2024
- **Member, Yonsei Electrical and Electronic Honor Society (EEHS) 42nd** [Yonsei University] — Mar. 2024

TEACHING & SERVICES

- **Tutor - Engineering Mathematics 1** [Yonsei EEHS] — Apr. - May. 2024
- **Mentor - Yonsei-Nexon $\sqrt{i}$ RC Creative Platform Peer Mentors** [Yonsei University, Nexon Korea] — Mar. – Dec. 2024
- **Tutor – High school Math, Incheon Bugwang High School** [Yonsei Social Engagement] — Mar. – Jun. 2023

AWARDS & HONORS

- **Selected Idea, K-Digital Challenge: NET Challenge Camp season 11** [National Information Society Agency] — Jul. 25. 2024
- **Excellence Award, 2023 Yonsei-Nexon $\sqrt{i}$ RC Creative Platform** [Yonsei University] — Nov. 29. 2023
- **Semester High Honors** [Yonsei University] — Fall 2024
- **Semester Honors** [Yonsei University] — Spring 2024, Spring 2023

SCHOLARSHIP

- **Presidential Science Scholarship** [Korea Student Aid Foundation] — 2025 - Present
  - ◦ Selected as one of 60 undergraduate students in STEM nationwide for the annual Korean Presidential Science Scholarship
- **Yoon Jayoung Emerging Innovator Scholarship** [Yonsei University] — 2024
- **Unhae Scholarship 11th** [Unhae Scholarship Foundation] — 2024
- **National Science and Engineering Scholarship** [Korea Student Aid Foundation] — 2023 – 2024

SKILLS

CAD	Altium Designer, Eagle CAD	**Programming**	MATLAB, Python, C/C++, Verilog
Circuit Simulation	Keysight ADS, PSpice, PLECS	**Hardware**	DAQ, Arduino, FPGA
EM Simulation	Ansys HFSS		